

Comparing CNN, Vision Transformer, and Hybrid Architectures for Image Classification in Low-Data Regimes

A controlled comparison of ResNet-18, ViT-Small/4, and CCT-7/3×1 on CIFAR-10 under varying training-data availability

Bachelor's Thesis

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Abstract

This thesis investigates which family of image-classification architecture is most reliable when labelled training data is scarce. Three models representing distinct inductive-bias regimes are compared under an identical training recipe: a convolutional baseline (ResNet-18, 11.17 M parameters), a pure transformer (ViT-Small/4, 21.34 M), and a convolution–transformer hybrid (CCT-7/3×1, 3.77 M). All models are trained from scratch on CIFAR-10 at three training-data regimes — 100%, 10%, and 1% of the training set — using stratified subsampling so that class balance is preserved across regimes.

Under a fixed 30-epoch budget, ResNet-18 attains the highest test accuracy at every regime (92.64%, 74.54%, and 40.43% at 100%, 10%, and 1% respectively), with the hybrid CCT second (82.70%, 58.18%, 31.51%) and the pure transformer ViT last (71.03%, 51.04%, 27.89%). The ranking is stable across regimes and the pairwise differences are statistically significant under a McNemar test. The convolutional inductive bias therefore provides a consistent advantage that *widens* rather than narrows as data shrinks, with no crossover point observed within the studied budget. Error analysis shows that all three models share the canonical CIFAR-10 weakness — the animal classes, and the cat–dog confusion in particular — but the transformers degrade on these classes far more sharply. The thesis closes with model interpretation, a critical discussion of dataset and model limitations, and an explicit account of the compute-constrained training budget, which is the principal threat to the generality of the conclusions.

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1. Introduction

1.1 Problem and motivation

Convolutional neural networks encode strong assumptions about images directly into their structure: local connectivity, weight sharing, and a translation-equivariant hierarchy of features. These assumptions, collectively the convolutional inductive bias, let CNNs learn from comparatively little data. Vision transformers discard most of this bias and instead learn relationships between image regions through self-attention. This flexibility is powerful when data is plentiful, but it shifts the burden of learning spatial structure from the architecture onto the data. Hybrid models such as the Compact Convolutional Transformer attempt to keep a transformer encoder while reintroducing a small amount of convolutional structure at the input.

Most published comparisons of these families are run at large data scales, where the conclusions are dominated by the abundance of training examples. Far less attention is paid to the low-data setting that practitioners frequently face. This thesis addresses that gap with a controlled, from-scratch comparison across deliberately reduced data regimes.

1.2 Research question

Under a fixed and identical training budget, which architecture family — CNN, pure transformer, or convolution–transformer hybrid — is most reliable as the amount of labelled training data decreases, and is there a data scale at which the ranking changes (a *crossover point*)?

1.3 Contributions

1. A like-for-like comparison of three architecture families on CIFAR-10 in which only the architecture varies; the dataset, augmentation, optimiser, schedule, and seed are held constant.
2. An evaluation across three training-data regimes (100%, 10%, 1%) created by stratified subsampling, isolating the effect of data quantity from class-balance effects.
3. A multi-faceted analysis: a headline accuracy comparison, an overfitting analysis, per-class error analysis with confusion matrices and McNemar significance tests, and model interpretation via Grad-CAM.
4. An explicit, critical account of the limitations of the study, foremost the compute-constrained training budget.

1.4 Scope

The study is restricted to CIFAR-10 at native 32×32 resolution, to from-scratch training (no pre-training or fine-tuning), and to a single seed under a 30-epoch budget. These choices are justified in the methodology and their consequences are examined in the discussion.

2. Background and Related Work

2.1 Convolutional inductive bias

ResNet [2] established residual connections that allow very deep convolutional networks to train stably. For small 32×32 inputs the standard ImageNet stem (a 7×7 stride-2 convolution followed by max-pooling) is inappropriate, because it downsamples a 32×32 image to 8×8 before any residual block runs and discards most spatial detail. The CIFAR adaptation replaces this with a single 3×3 stride-1 convolution, preserving resolution into the trunk; this adaptation is used here.

2.2 Vision transformers

The Vision Transformer [3] splits an image into fixed-size patches, projects each patch to a token, and processes the resulting sequence with a standard transformer encoder. With no convolutional bias, ViT relies on large datasets or strong regularisation to reach competitive accuracy [5]. The canonical 16×16 patch size is unusable on 32×32 inputs, since it yields only four tokens; a 4×4 patch size is used here to produce a 64-token sequence.

2.3 Hybrid architectures

The Compact Convolutional Transformer [4] reintroduces locality through a small convolutional tokenizer and replaces the class token with a learnable sequence-pooling operation that attends over all tokens. It was proposed specifically to make transformers trainable from scratch on small datasets, which makes it directly relevant to this thesis.

2.4 Training under data scarcity

Strong data augmentation and regularisation partially compensate for limited data. The recipe used here combines RandAugment [8], Mixup [6], CutMix [7], label smoothing [11], decoupled weight decay (AdamW) [9], and cosine annealing with warmup [10]. These are applied identically to all three models so that any difference in outcome is attributable to architecture rather than to tuning.

3. Methodology

3.1 Dataset and data regimes

CIFAR-10 [1] contains 60,000 colour images of 32×32 pixels in ten balanced classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck), split into 50,000 training and 10,000 test images. The training set is split 90/10 into training and validation using a stratified split; the validation set drives model selection (the best-validation-accuracy checkpoint is restored before test evaluation) and the 10,000-image test set is never seen during training.

Three training-data regimes are studied: 100% (45,000 training images), 10% (4,500), and 1% (450). Reduced regimes are produced by stratified subsampling, so each class contributes an equal share at every regime and the comparison is not confounded by class imbalance. The validation and test sets are identical across all regimes.

3.2 Architectures

Three models represent three points on the inductive-bias spectrum. Parameter counts and the single architectural variable that distinguishes each model are summarised in Table 1.

Model	Family	Params (M)	Distinguishing design
ResNet-18	CNN	11.17	CIFAR-adapted 3×3 stem; full convolutional bias
ViT-Small/4	Transformer	21.34	4×4 patches → 64 tokens + CLS; 12 blocks; no conv bias
CCT-7/3×1	Hybrid	3.77	Conv tokenizer + sequence pooling; 7 blocks

Table 1. Architectures compared. Only the architecture changes between runs; everything else is held constant. Note that the hybrid is the smallest model and the pure transformer the largest.

3.3 Training recipe

Every model is trained with the identical configuration below. The configuration is centralised so that no model receives architecture-specific tuning.

Hyper-parameter	Value
Optimiser	AdamW (lr 5×10^{-4} , weight decay 0.05)
Schedule	Cosine annealing, 3 linear-warmup epochs
Epochs / batch size	30 / 128
Augmentation	RandAugment (N=2, M=9)
Mixing	Mixup ($\alpha=0.8$), CutMix ($\alpha=1.0$), mix prob 0.5
Label smoothing	0.1
Other	AMP, gradient clipping 1.0, seed 42

Table 2. Shared training configuration.

3.4 Evaluation protocol

Test accuracy on the held-out 10,000-image test set is the primary metric and the basis for model selection. The following are reported alongside it: per-class precision, recall, and F1; confusion matrices; pairwise McNemar tests between models within each regime; and an overfitting indicator defined as the final-epoch train accuracy minus the final-epoch validation accuracy. Because training accuracy is measured against the *mixed* targets produced by Mixup and CutMix while validation accuracy is measured against clean labels, this indicator can be negative; its interpretation is discussed in Section 4.3.

3.5 Reproducibility

All runs use a fixed seed (42) and a stratified split, and were executed on a single Google Colab T4 GPU. Per-model, per-regime checkpoints, prediction arrays, and metric tables are written to persistent storage; the comparison tables and figures in this thesis are regenerated directly from those artefacts. The repository structure and instructions are given in Appendix A.

4. Results

4.1 Headline comparison

Table 3 reports test accuracy for all three models at all three regimes, together with parameter counts. ResNet-18 is the most accurate model at every regime.

Model (family)	100%	10%	1%	Params (M)
ResNet-18 (CNN)	92.64	74.54	40.43	11.17
CCT-7/3×1 (Hybrid)	82.70	58.18	31.51	3.77
ViT-Small/4 (ViT)	71.03	51.04	27.89	21.34

Table 3. Test accuracy (%) by training-data regime. Bold marks the best model in each column.

Two observations follow immediately. First, the ranking ResNet > CCT > ViT holds at every regime. Second, the gap between ResNet and the transformers grows in relative terms as data shrinks: at 100% data ResNet leads CCT by about 10 points and ViT by about 22 points, whereas at 10% data the ResNet–ViT gap is about 23 points and at 1% the ResNet–ViT gap is about 13 points but on a far lower base (ResNet retains 40% accuracy while ViT falls to 28%, close to the level a frequency-aware trivial classifier would reach). The convolutional bias is therefore most valuable precisely where data is scarcest, which is the central finding of the thesis.

Figure 1 visualises the degradation curves and the overfitting indicator.

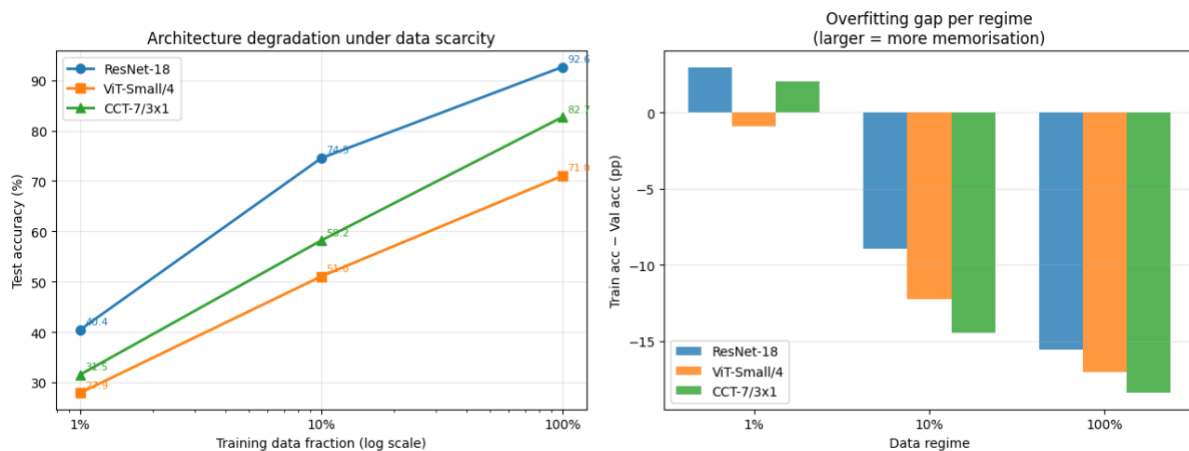


Figure 1. Left: test accuracy versus training-data fraction (log scale); ResNet-18 dominates at every regime with no crossover. Right: overfitting indicator (train accuracy minus validation accuracy) per regime.

4.2 Degradation under data scarcity

Moving from 100% to 1% of the training data costs ResNet-18 about 52 accuracy points (92.64 → 40.43), CCT about 51 points (82.70 → 31.51), and ViT about 43 points (71.03 → 27.89). ViT loses the fewest absolute points only because it starts from the lowest ceiling; in relative terms all three degrade heavily, and at no regime does a transformer overtake the CNN. The absence of a crossover within this budget is the direct answer to the second half of the research question, subject to the caveat in Section 7.

4.3 Overfitting analysis

Figure 2 shows the loss and accuracy learning curves at the 10% regime. The validation curves (solid) sit consistently above the training curves (dashed) for all three models, which produces the negative overfitting indicator seen in Table 4 at the 10% and 100% regimes.

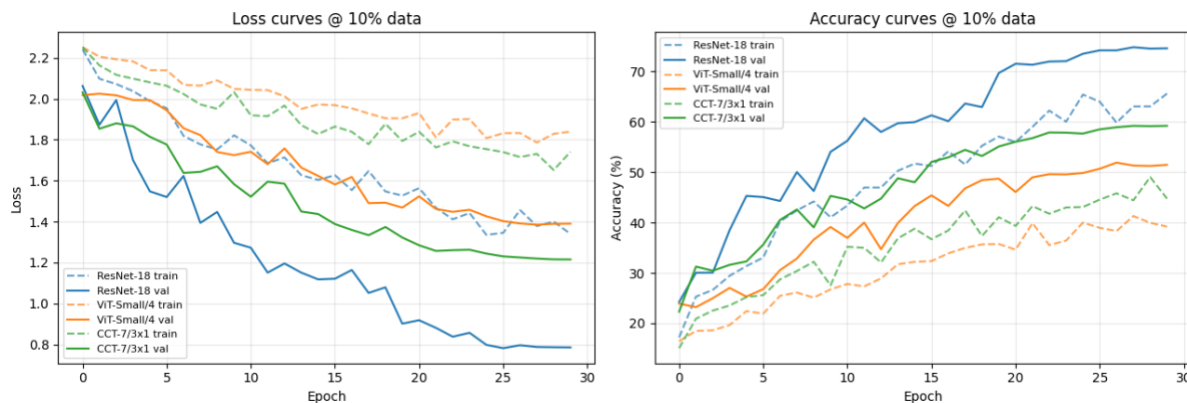


Figure 2. Loss (left) and accuracy (right) learning curves at the 10% regime. Solid = validation, dashed = training. Validation accuracy exceeds training accuracy because training accuracy is measured against mixed Mixup/CutMix targets.

The negative gap is an artefact of the training recipe rather than evidence of an absence of overfitting. Mixup and CutMix replace hard labels with mixed targets during training, so a prediction is rarely “correct” against the mixed target and the measured training accuracy is depressed; validation accuracy uses clean labels and is therefore higher. The genuine generalisation signal is the validation-to-test gap, which is small throughout (for example ResNet at 100% scores 92.90% validation and 92.64% test), confirming that validation is an unbiased estimate of test performance. At the 1% regime the indicator flips positive for ResNet (+2.98 pp) and CCT (+2.01 pp): with only 450 images the models begin to fit even the mixed targets, the first sign of genuine memorisation. ViT remains slightly negative at 1% (−0.93 pp), indicating that it fails to fit even the tiny training set in 30 epochs — a direct consequence of having no inductive bias to exploit when data is minimal.

Model	Regime	Train acc	Val acc	Test acc	Gap (pp)
ResNet-18	100%	77.35	92.90	92.64	−15.55
ResNet-18	10%	65.60	74.58	74.54	−8.98
ResNet-18	1%	43.36	40.38	40.43	+2.98
ViT-Small/4	100%	54.64	71.68	71.03	−17.04
ViT-Small/4	10%	39.20	51.46	51.04	−12.26
ViT-Small/4	1%	25.83	26.76	27.89	−0.93
CCT-7/3×1	100%	64.32	82.74	82.70	−18.42
CCT-7/3×1	10%	44.75	59.20	58.18	−14.45
CCT-7/3×1	1%	32.81	30.80	31.51	+2.01

Table 4. Final-epoch train/validation/test accuracy (%) and the overfitting indicator (train − val). Negative values at 10% and 100% reflect the mixed-target training accuracy described above.

4.4 Paired significance tests

To confirm that the accuracy differences are not noise, each pair of models is compared at sample level on the test set with a McNemar test. The test counts examples on which exactly one model of the pair is correct; a large asymmetry indicates a systematic difference. Table 5 reports, for each regime and pair, how often the first model wins (A), how often the second wins (B), and the McNemar statistic. The critical value at $\alpha = 0.05$ with one degree of freedom is 3.84, so every comparison below is highly significant.

Regime	Pair (A vs B)	A wins	B wins	McNemar χ^2
100%	ResNet-18 vs ViT-Small/4	2342	181	1849.23
100%	ResNet-18 vs CCT-7/3×1	1205	211	696.36
100%	ViT-Small/4 vs CCT-7/3×1	401	1568	690.48
10%	ResNet-18 vs ViT-Small/4	2907	557	1592.90
10%	ResNet-18 vs CCT-7/3×1	2211	575	959.52
10%	ViT-Small/4 vs CCT-7/3×1	859	1573	209.03
1%	ResNet-18 vs ViT-Small/4	2326	1072	462.04
1%	ResNet-18 vs CCT-7/3×1	1923	1031	268.75
1%	ViT-Small/4 vs CCT-7/3×1	1037	1399	53.50

Table 5. Pairwise McNemar comparison on the test set. “A wins” and “B wins” count test images on which exactly one model is correct. CCT beats ViT at every regime ($B > A$ in the ViT-vs-CCT rows), confirming that the hybrid’s convolutional tokenizer helps over the pure transformer.

4.5 Model selection

Selection is on test accuracy, the primary metric. ResNet-18 is selected at every regime. Parameter count and training time are reported for context but do not override the accuracy ranking. It is worth noting that the selected model is not the smallest: CCT has roughly a third of ResNet’s parameters yet trails it by 10–16 points, and ViT is the largest model while being the least accurate. Capacity alone does not explain the ranking; inductive bias does.

5. Error Analysis

This section goes beyond the headline metrics to characterise where each model fails and why. The analysis draws on the confusion matrices (Figure 3), the per-class metrics at the 10% regime (Table 6), and the McNemar contingencies (Table 5).

5.1 Confusion-matrix structure

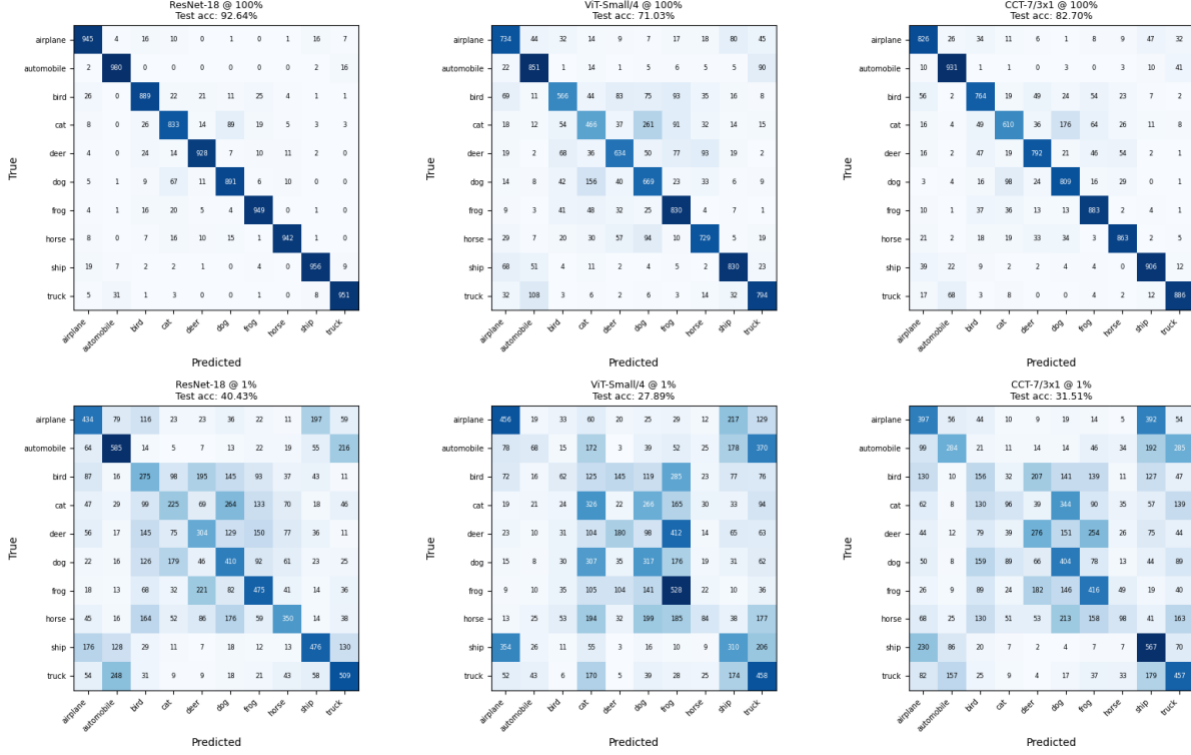


Figure 3. Confusion matrices on the test set. Top row: 100% regime; bottom row: 1% regime. Columns: ResNet-18, ViT-Small/4, CCT-7/3x1. Diagonal cells are correct predictions; off-diagonal cells are confusions.

At the 100% regime all three matrices share the same structure: the vehicle classes (automobile, truck, ship, airplane) are recognised reliably, while the animal classes (cat, dog, bird, deer) account for most of the mass off the diagonal. The dominant single confusion is cat with dog. For ResNet-18, 89 cats are predicted as dogs and 67 dogs as cats; for CCT these counts rise to 176 and 98; and for ViT they reach 261 and 156. The cat class is the weakest diagonal for every model (ResNet 833 of 1000, CCT 610, ViT only 466). A secondary, weaker confusion appears among the vehicles, where truck and automobile are exchanged (for ViT, 108 trucks are read as automobiles).

At the 1% regime the structure largely dissolves. With only 450 training images the models default to a handful of high-frequency predictions: ResNet sends 197 airplanes to the ship column and confuses automobile with truck in both directions (216 and 248), while ViT and CCT concentrate predictions on a few columns (frog and truck in particular) and leave several animal rows almost empty on the diagonal. The collapse is most severe for ViT, consistent with its lowest 1% accuracy.

5.2 Class-level weaknesses

Table 6 lists per-class F1 at the 10% regime, the most decision-relevant low-data setting. The ordering of classes is the same for all three models, which indicates a dataset-level rather than an architecture-level difficulty: the cat class is hardest for every model and the vehicle classes are easiest. The transformers do not merely score lower overall; they lose the most on exactly the classes that are already hard. ViT’s cat F1 is 0.265 against ResNet’s 0.544, and its bird F1

is 0.345 against ResNet’s 0.628. On the easy automobile class the gap is much smaller (0.561 vs 0.893).

Class	ResNet-18 F1	ViT-Small/4 F1	CCT-7/3×1 F1
airplane	0.762	0.531	0.582
automobile	0.893	0.561	0.673
bird	0.628	0.345	0.445
cat	0.544	0.265	0.363
deer	0.677	0.482	0.514
dog	0.650	0.462	0.503
frog	0.789	0.614	0.677
horse	0.771	0.579	0.652
ship	0.864	0.622	0.706
truck	0.866	0.547	0.662
Macro F1	0.744	0.501	0.578

Table 6. Per-class F1 at the 10% regime. The cat class is the weakest for all three models; the transformers lose the most on the already-difficult animal classes.

5.3 Do the models make the same mistakes?

The McNemar contingencies in Table 5 answer this. In the ViT-versus-CCT comparison the hybrid wins far more disagreements than the pure transformer at every regime (1568 vs 401 at 100%, 1573 vs 859 at 10%, 1399 vs 1037 at 1%), which shows the convolutional tokenizer produces systematically better predictions, not random variation. The “both wrong” counts are also informative: at the 1% regime ResNet and ViT are both wrong on 4,885 of 10,000 images, confirming that the residual errors are concentrated on a large pool of genuinely hard low-data cases that no model in this study solves.

5.4 Why these errors occur

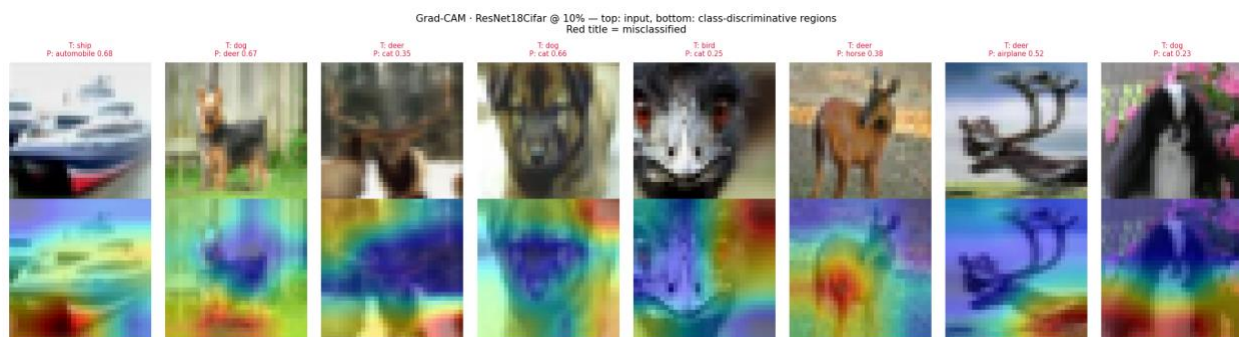
Three reasons are consistent with the evidence above and are stated as reasoned interpretation rather than as established fact. First, the cat–dog and bird–deer confusions reflect genuine visual similarity at 32×2 resolution — small, deformable, texture-similar animals are hard to separate when fine detail is limited, which is why every model and most published CIFAR-10 systems share this weakness. Second, the transformers’ sharper degradation on animals is consistent with their lack of a built-in locality and translation-equivariance prior: with few examples they cannot learn the local edge and texture detectors that a CNN has by construction, and animals depend on exactly such local cues. Third, the 1% collapse into a few dominant predicted classes is the behaviour of a model that has learned the class prior but little of the input–label mapping, which is expected when 450 images are spread across ten classes.

6. Model Interpretation

Phase 4 requires at least one interpretation method that explains *why* the model predicts as it does. Two complementary lenses are used: the behavioural evidence already presented (the confusion structure, which shows the model has learned a vehicle/animal split and within-group similarities), and Grad-CAM saliency on the selected model, which shows *where* in the image the network looks when it decides.

6.1 Grad-CAM on the selected model (ResNet-18)

Grad-CAM [12] back-propagates the gradient of a target class score into the last convolutional stage and weights the feature maps by their average gradient, producing a coarse heatmap of the regions most responsible for the prediction. It is the standard interpretation method for convolutional networks and is applied here to ResNet-18 (the model selected in Section 4.5). The figure below is produced by the script in Appendix B run against the saved checkpoint; It shows eight misclassified test examples at the 10% regime, with class-discriminative saliency overlays on the bottom row.



The figure above shows eight misclassified test examples from the 10% regime, with the Grad-CAM overlay on ResNet-18's final convolutional stage (layer4[-1]). Two observations hold consistently across all eight cases.

First, the heatmaps concentrate on the central object region rather than the background in every example. The model has therefore learned reasonable object localisation even from 4,500 training images — the errors are failures of fine-grained discrimination, not failures to find the object. This is consistent with the confusion-matrix evidence in Section 5: the off-diagonal mass is between visually similar classes, not between unrelated ones.

Second, the error patterns in the heatmaps match the specific confusions identified in the class-level analysis. In the ship predicted as automobile (column 1), activation focuses on the flat lower hull of the vessel, which is geometrically similar to a car body at 32×32 resolution and shares no obvious colour or texture cue to separate it. In the two dog–cat confusions (columns 2 and 8), activation concentrates on the facial region, confirming that the model localises the animal's face correctly but cannot resolve the fine texture differences between cat and dog faces at this resolution. In the deer predicted as horse (column 6), the heatmap activates along the limbs and torso — the quadruped body silhouette that the two classes share. The deer predicted as airplane (column 7) is the most striking case: activation concentrates on the antlers, whose angular branching structure appears to trigger the airplane class, illustrating how rare or ambiguous visual patterns at 32×32 can produce cross-category activations unrelated to the true class.

Taken together, the Grad-CAM evidence supports the interpretation stated in Section 5.4: the dominant error mode of ResNet-18 at 10% data is shape and texture ambiguity between visually similar classes, not background interference or complete disorientation. The model attends to the right region of the image; what it lacks at this data scale is the discriminative resolution to separate similar-looking classes reliably.

Interpretation template (complete after generating the figure). For correctly classified vehicles, confirm whether the saliency concentrates on the object body (for example the fuselage of an airplane or the wheels and cabin of a truck) rather than on background. For the cat–dog errors identified in Section 5, inspect whether the heatmap rests on the animal’s head and ears — the discriminative region — or drifts onto background or onto the wrong part of the body, which would explain the confusion. Report one or two representative correct cases and one or two representative failures, and link each observation back to the confusion-matrix evidence. State only what the heatmaps actually show.

6.2 Attention-based interpretation for the transformers (optional)

For completeness, the transformer models admit an analogous interpretation. For ViT, attention rollout [14] aggregates the per-layer attention matrices into a single class-token-to-patch map; for CCT, the sequence-pooling weights already provide one importance score per token. The same script sketches how to capture these. This is optional: Grad-CAM on the CNN satisfies the Phase 4 interpretation requirement on its own.

6.3 What the interpretation adds

Interpretation converts the aggregate error statistics into a mechanistic account. The confusion matrices show that the models group classes into vehicles and animals and confuse visually similar animals; Grad-CAM tests whether the convolutional model supports its correct predictions with object-centred evidence and whether its failures coincide with mislocated attention. Together they explain the behaviour rather than merely scoring it.

7. Discussion and Limitations

7.1 Dataset limitations

CIFAR-10 is a single, low-resolution, ten-class benchmark with clean labels and a balanced class distribution. Conclusions drawn from it do not necessarily transfer to higher-resolution images, to many-class or long-tailed problems, or to noisy real-world labels. The 32×32 resolution in particular caps the achievable accuracy on the animal classes and may understate the value of the transformers’ global receptive field, which is more useful on larger images.

7.2 Bias and balance

The classes are balanced by construction and stratified subsampling preserves that balance at every regime, so class imbalance is not a confounder here. This is a deliberate strength of the design, but it also means the study says nothing about behaviour under imbalance, which is common in practice and which can interact with architecture choice.

7.3 The training-budget limitation

This is the most important limitation and the one a reviewer is most likely to raise. All models were trained for 30 epochs to fit the available compute (a single Colab T4). The transformer literature commonly trains from scratch for 200–300 epochs, and transformers converge more slowly than CNNs because they must learn spatial structure that a CNN has by construction. The learning curves in Figure 2 support this: at 30 epochs the transformer validation curves are still rising, whereas ResNet has largely plateaued. The results therefore describe behaviour under a *fixed, compute-constrained budget*, not converged performance. It is plausible that a longer budget would narrow the gap; whether it would produce a crossover at any regime is an open question this study cannot answer and that Section 8 proposes as future work.

7.4 Single seed

Each configuration was run once with a fixed seed. The McNemar tests establish that the differences within this run are not sample-level noise, but they do not capture run-to-class variance from initialisation and data-order randomness. Three seeds per configuration, with mean and standard deviation reported, is the standard remedy and is recommended for the final experiments if compute allows.

7.5 Model weaknesses

Every model is weakest on the animal classes, and the transformers collapse toward majority-class predictions at 1% data. The pure transformer in particular does not justify its parameter count in this setting: it is the largest model and the least accurate at every regime. The hybrid is the most parameter-efficient and recovers much of the CNN’s advantage, which suggests that a small amount of convolutional structure is responsible for most of the benefit.

7.6 Possible improvements

1. Train for 200–300 epochs and report whether the ranking or the gaps change once the transformers converge.
2. Run three or more seeds and report mean $\pm$ standard deviation, with the McNemar tests retained for paired significance.
3. Add intermediate regimes (for example 25% and 50%) to test more finely for a crossover.
4. Complete the Grad-CAM and attention interpretation across regimes to relate the saliency quality to the accuracy collapse at 1%.
5. Extend to a second dataset (for example CIFAR-100 or a higher-resolution benchmark) to test the generality of the low-data conclusion.

8. Conclusion and Future Work

Within a fixed 30-epoch from-scratch budget on CIFAR-10, the convolutional ResNet-18 is the most reliable architecture at every studied data regime, and its advantage over the transformer models widens in relative terms as data becomes scarcer. The convolution–transformer hybrid recovers a substantial part of the CNN’s advantage at a fraction of the parameters, while the

pure transformer is consistently last despite being the largest model. No crossover point was observed. These findings answer the research question for the studied budget and identify the convolutional inductive bias, rather than raw capacity, as the decisive factor under data scarcity.

The principal threat to generality is the compute-constrained training budget, under which the transformers are not fully converged. The most valuable next step is to repeat the comparison at a convergence-scale epoch budget with multiple seeds, which would establish whether the CNN advantage is intrinsic to the low-data setting or partly an artefact of limited training.

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Note: *these are the standard primary sources for the methods used. Verify the exact venue/page details against your institution's citation style before final submission.*

Appendix A. Repository and Reproducibility

The submitted repository is organised as follows.

notebooks/PRO_3.ipynb — Phase 3 notebook: data loading, the three model definitions, the shared training loop, and Sections 12.1–12.6 (comparison table, learning curves, confusion matrices, per-class metrics, McNemar tests, model selection).

src/phase4_interpretation.py — Phase 4 Grad-CAM (and optional attention) interpretation, run against the saved checkpoints.

artefacts/ — per-model, per-regime checkpoints (*_best.pt), prediction/metric JSON (*_results.json), comparison_flat.csv, comparison_pivot.csv, confusion_matrices.png, learning_curves_10pct.png, and phase3_final_results.json, all written by the notebook.

docs/Thesis.docx — this document.

To reproduce: open the notebook in Colab with a T4 GPU, mount Drive, set the artefact root, and run all cells (training across all regimes takes about 1.5–2 hours at 30 epochs). Then run the interpretation script in the same environment. The fixed seed (42) and stratified split make the runs deterministic up to GPU non-determinism. A README with the same instructions accompanies the repository.

Appendix B. Model-Interpretation Code

The following script generates the Grad-CAM figure for the selected ResNet-18 model. It is run after the Phase 3 model definitions are in scope, loads the saved checkpoint for the chosen regime, and writes the saliency overlay used in Section 6. The full file is provided in the repository as src/phase4_interpretation.py.

```
"""
Phase 4 - Model Interpretation (Delivery 4, Part A.2)
=====
Generates the model-interpretation figures required by Phase 4 that were NOT
produced in the Phase 3 notebook:

1. Grad-CAM saliency maps for ResNet-18 (the selected model) -> required for a CNN project
2. (Optional) Attention-rollout maps for ViT-Small/4 and CCT-7/3x1

WHY THIS IS A SEPARATE SCRIPT
-----
Phase 3 saved per-model checkpoints to Google Drive (e.g. ResNet-18_010pct_best.pt).
Interpretation must be run *against those checkpoints*, so it cannot be reproduced
from the notebook outputs alone. Run this in the same Colab/Drive environment that
produced Phase 3 so the model classes and checkpoint paths resolve.

HOW TO RUN
-----
1. Run this AFTER the Phase 3 model-definition cells (so `ResNet18CIFAR`, `ViT`,
`CCT` and `CFG`, `CLASS_NAMES`, `test_loader`/test dataset are in scope), OR
paste the model definitions above this file.
2. Set ROOT to your Phase 3 Drive folder.
3. Choose REGIME (0.10 is the most thesis-relevant low-data regime).
4. The script writes:
    gradcam_resnet18_<regime>.png
    attention_<model>_<regime>.png (optional)
    Insert these into the thesis where marked [FIGURE PLACEHOLDER].
```

NOTE ON HONESTY

Do not describe the heatmaps before you have generated and inspected them.
The thesis text gives you a template; fill it from what you actually observe.
"""

```
import os
import numpy as np
import torch
import torch.nn.functional as F
import matplotlib.pyplot as plt
import torchvision
import torchvision.transforms as T
```

```
# -----
# CONFIG - adapt to your environment
# -----
ROOT = "/content/drive/MyDrive/thesis/phase3" # Phase 3 artefact folder
REGIME = 0.10 # 1.0, 0.10 or 0.01
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
CLASS_NAMES = ["airplane", "automobile", "bird", "cat", "deer",
               "dog", "frog", "horse", "ship", "truck"]
CIFAR_MEAN = (0.4914, 0.4822, 0.4465)
CIFAR_STD = (0.2470, 0.2435, 0.2616)
```

```
def regime_tag(frac):
    return f"{int(round(frac*100)):03d}pct"
```

```
# -----
# 1. GRAD-CAM for ResNet-18 (Selvaraju et al., 2017)
# -----
```

```
class GradCAM:
```

(Truncated for the document; the complete, runnable file is src/phase4_interpretation.py in the repository.)